Healthcare Revenue Cycle & Operational Intelligence Analysis Using SQL

Main Objective

The main goal of this project is to analyze healthcare claim lifecycle efficiency, financial flow blockage, provider performance, and operational channel productivity using SQL-based business intelligence.

Business Problem

Healthcare organizations face operational challenges such as:

- Delayed claim payments
- Revenue stuck in workflow stages
- Poor provider documentation quality
- Differences in submission channel performance
- Financial leakage due to processing delay

Hospitals want:

- Faster cash flow
- Better claim management
- Operational efficiency

Dataset Meaning

This dataset represents healthcare operational intelligence including:

- Claim lifecycle tracking
- Financial exposure monitoring
- Patient demographic behaviour
- Provider performance quality
- Submission channel efficiency

Data Scope Note

This dataset is a structured sample dataset used to demonstrate healthcare revenue cycle analytics logic.

The purpose of this project is to showcase SQL-based business intelligence techniques such as lifecycle analysis, financial blockage evaluation, provider performance measurement, and channel productivity assessment.

The analytical framework demonstrated here can be directly applied to enterprise-scale healthcare datasets.

Why This Project Matters in Modern Healthcare

Healthcare industry is rapidly moving toward digital and intelligent claim processing systems. Hospitals are focusing on faster revenue cycle management and operational transparency.

This project is aligned with modern healthcare transformation where data-driven decision making is becoming important.

Digital healthcare platforms are increasing claim submission automation and reducing manual processing errors.

Revenue cycle automation is a major trend in 2026 healthcare technology.

Operational intelligence analysis helps hospital management understand workflow efficiency.

AI-assisted claim verification is growing but business strategy and financial optimization are still controlled by human decision systems.

This project focuses on business intelligence using SQL analytics.

Database Optimization

Indexing is used in this project to improve query performance.

Since filtering is frequently done on Claim Status and Provider ID, index creation helps faster data retrieval.

Indexes reduce database search time and improve system efficiency.

View creation is used to build a reporting layer for business visualization.

The view helps connect analytical queries with dashboard tools.

Business Value Impact of This Analysis

This analysis provides important operational and financial insights for hospital decision makers.

It helps reduce claim processing delay by identifying workflow bottlenecks.

Financial blockage analysis shows revenue locked in pending claim stages.

Provider performance analysis helps identify documentation quality issues.

Channel efficiency analysis supports digital healthcare adoption.

Risk segmentation helps hospital financial planning.

Business Problems Addressed

This project answers important operational healthcare questions.

It helps identify where claims are getting delayed in lifecycle stages.

Provider approval performance is analyzed to check documentation quality.

Submission channel efficiency is compared between paper, online, and phone methods.

Revenue distribution across claim workflow stages is measured.

Patient demographic behaviour is also studied.

Query Output Screenshot

1. Claim Lifecycle Efficiency Analysis

How are claims distributed across lifecycle stages, and where do volume concentrations indicate potential processing bottlenecks?

```
SELECT
    ClaimStatus,
    AgeGroup,
    CleanPatientAge,
    AnomalyFlag,
    COUNT(*) AS TotalClaims,

    COUNT(*) * 100.0 / SUM(COUNT(*)) OVER() AS Percentage

FROM ClaimData
GROUP BY
    ClaimStatus,
    AgeGroup,
    CleanPatientAge,
    AnomalyFlag

ORDER BY TotalClaims DESC;
```

Output:

ClaimStatus	AgeGroup	CleanPatientAge	AnomalyFlag	TotalClaims	Percentage
Approved	Adult	57	0	25	0.55556
Pending	Senior	81	0	24	0.53333
Approved	Senior	97	0	24	0.53333
Denied	Senior	70	0	24	0.53333
Approved	Young Adult	22	0	23	0.51111
Approved	Adult	51	0	23	0.51111
Denied	Young Adult	25	0	23	0.51111
Denied	Senior	76	0	22	0.48889
Approved	Young Adult	24	0	22	0.48889
Denied	Senior	74	0	22	0.48889
Pending	Senior	87	0	21	0.46667

- **Insight:** Claims are fairly evenly spread across stages like Approved, Pending, and Denied, with each category handling around 22 to 25 claims.
- The high concentration of Pending claims for Seniors suggests a potential processing bottleneck in verifying or approving claims for this specific age group.

2. Financial Blockage Analysis

What is the revenue value associated with non-finalized claim stages?

```
SELECT
ClaimStatus,
SUM(ClaimAmount) AS TotalBlockedRevenue
FROM ClaimOperationalView
GROUP BY ClaimStatus;
```

Output:

ClaimStatus	TotalBlockedRevenue
Approved	7798110.15
Denied	7550894.18
Pending	7214913.07

- **Insight:** There is approximately 7.2 million in revenue currently "blocked" or tied up in the Pending stage, which represents the non-finalized claims.
- The revenue is distributed relatively evenly across all three statuses, indicating a high volume of financial value moving through each phase of the claim lifecycle.

3. Provider Performance Analysis

Are there providers associated with high rejection or non-approved claim patterns?

```
SELECT

ProviderID,

COUNT(*) AS TotalClaims,

SUM(CASE WHEN ClaimStatus='Approved' THEN 1 ELSE 0 END)*100.0/COUNT(*)
AS ApprovalRate

FROM ClaimData

GROUP BY ProviderID;
```

Output:

ProviderID	TotalClaims	ApprovalRate
001333d4-0aba-4805-87b5-49f4c88d69ca	1	0.00000
002882c5-9ea4-4158-bef7-2a046c6f7ecb	1	0.00000
0038f9e8-145e-44df-bd43-4e26ff327e58	1	0.00000
004d07a1-c7ae-485f-a2ca-feb99c30f176	1	0.00000
00698dd9-fb4c-4365-a82b-8cf0c67482e9	1	0.00000
006c8459-efb3-4a9b-ba66-b288c7b937a8	1	100.00000
006ed7e1-155a-4f74-812e-ce02188734ea	1	100.00000
00798418-a57a-4e65-83bd-eeab0fd3a971	1	0.00000
00828d51-ec50-4df6-b2ce-6db19c3eceec	1	0.00000
008814ca-f16a-411d-98f3-436c71e9254c	1	0.00000
008e7848-d15a-4715-b73c-040ef678b038	1	0.00000

- **Insight:** Most providers shown have an ApprovalRate of 0%, meaning their claims are consistently being rejected or not approved.
- The data currently only shows one claim per provider, so more data is needed to confirm if these are long-term patterns or just isolated incidents.

4. Demographic Behaviour Pattern

How is claim status distributed across different patient age groups?

```
SELECT

    AgeGroup,

    ClaimStatus,

    COUNT(*) AS TotalClaims,

    COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(PARTITION BY AgeGroup) AS
PercentageWithinAgeGroup

FROM ClaimData

GROUP BY AgeGroup, ClaimStatus

ORDER BY AgeGroup, TotalClaims DESC;
```

Output:

AgeGroup	ClaimStatus	TotalClaims	PercentageWithinAgeGroup
Adult	Approved	392	35.66879
Adult	Denied	374	34.03094
Adult	Pending	333	30.30027
Child	Denied	261	34.52381
Child	Pending	255	33.73016
Child	Approved	240	31.74603
Senior	Pending	610	34.02119
Senior	Denied	610	34.02119
Senior	Approved	573	31.95761
Unknown	Approved	16	35.55556
Unknown	Pending	15	33.33333

Insight: Across all age groups, claims are distributed almost equally, with each status (Approved, Denied, Pending) making up roughly one-third of the total.

The Senior group has the highest volume of claims, while Adults have the highest individual approval rate at approximately 35.7%.

5. Channel Efficiency Analysis

How does claim approval performance vary across different submission methods?

```
SELECT

ClaimSubmissionMethod,

COUNT(*) AS TotalClaims,

SUM(CASE WHEN ClaimStatus='Approved' THEN 1 ELSE 0 END)*100.0/COUNT(*)
AS ApprovalRate

FROM ClaimData

GROUP BY ClaimSubmissionMethod;
```

Output:

ClaimSubmissionMethod	TotalClaims	ApprovalRate
Paper	1544	33.93782
Online	1461	32.85421
Phone	1495	34.64883

- **Insight:** All submission methods (Paper, Online, Phone) have very similar approval rates, ranging closely between 32.8% and 34.6%.
- Paper submissions have the highest volume of claims, while Phone submissions show the slightly highest success rate for approvals.

6. Risk Segmentation Analysis

What is the volume distribution of claims across low, medium, and high claim amount risk bands?

```
SELECT

CASE

WHEN ClaimAmount < 2000 THEN 'Low'

WHEN ClaimAmount BETWEEN 2000 AND 8000 THEN 'Medium'

ELSE 'High'

END AS ClaimSegment,

COUNT(*) AS TotalClaims

FROM ClaimData

GROUP BY ClaimSegment;
```

Output:

ClaimSegment	TotalClaims
Low	902
Medium	2696
High	902

Insight: The vast majority of claims fall into the Medium risk band, which has three times more claims than the Low or High categories.

The Low and High risk bands have an identical number of claims, showing a perfectly balanced distribution at both extremes of the claim amounts.

7. Specialty Performance Analysis

What is the distribution of claim lifecycle status across provider specialties?

```
SELECT
    ProviderSpecialty,
    ClaimStatus,
    COUNT(*) AS ClaimCount,
    COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(PARTITION BY ProviderSpecialty) AS PercentageWithinSpecialty
FROM ClaimData
GROUP BY ProviderSpecialty, ClaimStatus
ORDER BY ProviderSpecialty, ClaimCount DESC;
```

Output:

ProviderSpecialty	ClaimStatus	ClaimCount	PercentageWithinSpecialty
Cardiology	Denied	312	34.39912
Cardiology	Approved	311	34.28886
Cardiology	Pending	284	31.31202
General Practice	Denied	312	35.45455
General Practice	Pending	288	32.72727
General Practice	Approved	280	31.81818
Neurology	Approved	307	35.49133
Neurology	Denied	296	34.21965
Neurology	Pending	262	30.28902
Orthopedics	Approved	325	36.39418
Orthopedics	Pending	297	33.25868

- **Insight:** Across all specialties, claims are split almost equally (around 30-36%) between Approved, Denied, and Pending statuses.
- Orthopedics shows the highest individual approval rate at 36.4%, while General Practice has the highest denial rate at 35.5%.

8. Productivity Analysis

Which submission method handles higher claim volume and value on average?

```
SELECT
    ClaimSubmissionMethod,
    COUNT(*) AS TotalClaims,
    AVG(ClaimAmount) AS AvgClaimValue,
    COUNT(*) * 100.0 / SUM(COUNT(*)) OVER() AS VolumeShare
FROM ClaimData
GROUP BY ClaimSubmissionMethod
ORDER BY TotalClaims DESC;
```

Output:

ClaimSubmissionMethod	TotalClaims	AvgClaimValue	VolumeShare
Paper	1544	5060.961988	34.31111
Phone	1495	5019.197425	33.22222
Online	1461	4959.679630	32.46667

- **Insight:** Paper submissions lead in both total volume and average claim value, making it the primary method for high-value claims.
- The distribution is very balanced, with all three methods (Paper, Phone, Online) each handling roughly one-third of the total claim volume.

Recommendations After Analysis

Operational Improvement Suggestions

- Hospitals should promote digital claim submission systems.
- Providers with low approval rates should be audited or trained.
- Pending claim financial risk should be monitored regularly.
- Medical documentation quality should be improved.
- AI-based claim validation pipeline can be developed in future.